

1 Introduction to Discrete Mathematics

Discrete Mathematics (DM) is the study of mathematical structures that are fundamentally discrete (countable, distinct) rather than continuous (like real numbers in calculus). It deals with objects that can be enumerated, such as integers, graphs, and sets. DM is crucial for computer science, as it underpins algorithms, data structures, cryptography, logic circuits, databases, and more. Unlike continuous math (e.g., calculus), DM focuses on finite processes, proofs, and combinatorial reasoning.

1.1 Key Characteristics

- **Discrete vs. Continuous:** DM handles finite or countably infinite sets (e.g., integers), while continuous math deals with uncountable sets (e.g., real numbers).
- **Core Themes:** Logic, sets, relations, counting, graphs, and proofs.
- **History:** Roots in ancient logic (Aristotle) and combinatorics (Pascal), formalized in the 20th century by mathematicians like Gödel and Turing.
- **Applications:** Algorithm design, software verification, AI (e.g., decision trees), networking (graphs), and cryptography (modular arithmetic).

DM is often a prerequisite for advanced CS topics. Learning it builds problem-solving skills through rigorous proofs and abstract thinking.

2 Logic and Proofs

Logic is the foundation of DM, enabling reasoning and proofs.

2.1 Propositional Logic

- **Propositions:** Statements that are true (T) or false (F), e.g., “It is raining” (p).
- **Connectives:**
 - Negation ($\neg p$): Not p .
 - Conjunction ($p \wedge q$): p AND q (true if both true).
 - Disjunction ($p \vee q$): p OR q (true if at least one true).
 - Implication ($p \rightarrow q$): If p then q (false only if p true and q false).
 - Biconditional ($p \leftrightarrow q$): p if and only if q (true if both same).
- **Truth Tables:** Evaluate compound statements.

Example Truth Table for $(p \wedge q) \rightarrow r$:

- **Tautology:** Always true (e.g., $p \vee \neg p$).
- **Contradiction:** Always false (e.g., $p \wedge \neg p$).
- **Logical Equivalence:** Statements with same truth values (e.g., $\neg(p \wedge q) \equiv \neg p \vee \neg q$, De Morgan’s Law).

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

Table 1: Truth Table for $(p \wedge q) \rightarrow r$

2.2 Predicate Logic

- Extends propositional logic with quantifiers.
- **Universal** ($\forall$): For all (e.g., $\forall x \in \mathbb{N}, x \geq 0$).
- **Existential** ($\exists$): There exists (e.g., $\exists x \in \mathbb{R}, x^2 = 2$).
- **Negation**: $\neg \forall x P(x) \equiv \exists x \neg P(x)$.

2.3 Proof Techniques

- **Direct Proof**: Assume premise, derive conclusion (e.g., prove if n even, then n^2 even: $n = 2k \rightarrow n^2 = 4k^2$ even).
- **Contrapositive**: Prove $\neg q \rightarrow \neg p$ instead of $p \rightarrow q$.
- **Contradiction**: Assume negation, derive absurdity.
- **Induction**: Base case + inductive step (e.g., prove sum 1 to $n = \frac{n(n+1)}{2}$).
 - Base: $n = 1, 1 = \frac{1(2)}{2} = 1$.
 - Assume for k : sum = $\frac{k(k+1)}{2}$.
 - For $k + 1$: sum + $(k + 1) = \frac{k(k+1)}{2} + (k + 1) = \frac{(k+1)(k+2)}{2}$.
- **Pigeonhole Principle**: If n items into m containers ($n > m$), at least one container has > 1 item.

3 Set Theory

Sets are collections of distinct objects.

3.1 Basics

- **Notation**: $\{a, b, c\}$, $\emptyset$ (empty set), $\mathbb{N}$ (naturals), $\mathbb{Z}$ (integers), $\mathbb{R}$ (reals).
- **Operations**:
 - Union ($A \cup B$): Elements in A or B.
 - Intersection ($A \cap B$): Elements in both.
 - Difference ($A - B$): In A but not B.
 - Complement (A^c): Not in A (relative to universe).

- Cartesian Product ($A \times B$): Pairs (a, b) .
- **Subsets** ($\subseteq$): All elements of A in B. Power set (2^A): All subsets of A (size $2^{|A|}$).
- **Cardinality** ($|A|$): Number of elements.

3.2 Laws

- De Morgan's: $(A \cup B)^c = A^c \cap B^c$.
- Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$.

Example: Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$. Then $A \cup B = \{1, 2, 3, 4\}$, $|A \cup B| = 4$.

4 Relations and Functions

4.1 Relations

- **Definition:** Subset of $A \times B$ (pairs relating elements).
- **Properties:**
 - Reflexive: (a, a) for all a .
 - Symmetric: If (a, b) then (b, a) .
 - Transitive: If (a, b) and (b, c) then (a, c) .
- **Equivalence Relation:** Reflexive, symmetric, transitive (partitions into classes, e.g., congruence modulo n).
- **Partial Order:** Reflexive, antisymmetric, transitive (e.g., $\leq$ on integers).

4.2 Functions

- **Definition:** Relation where each a in domain maps to exactly one b in codomain ($f : A \rightarrow B$).
- **Types:**
 - Injective (one-to-one): Different inputs $\rightarrow$ different outputs.
 - Surjective (onto): Every codomain element hit.
 - Bijective: Both (invertible).
- **Composition:** $(f \circ g)(x) = f(g(x))$.
- **Inverse:** For bijective f , f^{-1} undoes f .

Example: $f(x) = 2x$ (injective on $\mathbb{R}$, but not surjective onto $\mathbb{R}$ as misses negatives).

5 Combinatorics

Counting techniques for discrete objects.

5.1 Basic Counting

- **Rule of Sum/Product:** Choices add/multiply.
- **Permutations:** Arrangements ($P(n, k) = \frac{n!}{(n-k)!}$).
- **Combinations:** Selections ($C(n, k) = \frac{n!}{k!(n-k)!}$).
- **Binomial Theorem:** $(x + y)^n = \sum C(n, k)x^{n-k}y^k$.

5.2 Advanced

- **Inclusion-Exclusion:** Generalizes for multiple sets.
- **Generating Functions:** Polynomials for counting (e.g., $(1 + x + \frac{x^2}{2!} + \dots)$ for exponentials).
- **Recurrence Relations:** Define sequences (e.g., Fibonacci: $F_n = F_{n-1} + F_{n-2}$, $F_0 = 0$, $F_1 = 1$).
 - Solving: Characteristic equation or generating functions.

Example: Number of ways to choose 3 fruits from 5 apples and 3 oranges (with types): $C(5, 3) + C(3, 3) +$ mixed terms, but for distinct, use multisets.

6 Graph Theory

Graphs model relationships.

6.1 Basics

- **Graph** $G = (V, E)$: Vertices (V), edges (E).
- **Types:** Undirected/directed, weighted/unweighted, simple/multigraph.
- **Degree:** Number of edges at vertex (handshaking lemma: $\sum \text{degrees} = 2|E|$).
- **Paths/Cycles:** Sequence of edges; cycle returns to start.
- **Connected Graph:** Path between any vertices.
- **Trees:** Acyclic connected graphs ($|E| = |V| - 1$).

6.2 Key Concepts

- **Eulerian Path/Circuit:** Traverses each edge once (path: even degrees except 0/2; circuit: all even).
- **Hamiltonian Path/Circuit:** Visits each vertex once (NP-complete to detect).
- **Planar Graphs:** Drawable without crossings (Euler's formula: $V - E + F = 2$ for faces).
- **Coloring:** Assign colors so no adjacent same (chromatic number).

6.3 Algorithms

- **Traversal:** DFS (stack/recursion), BFS (queue).
- **Shortest Path:** Dijkstra (non-negative weights), Bellman-Ford (handles negatives).
- **MST:** Kruskal (sort edges, union-find), Prim (greedy from vertex).

Example: In a complete graph K_n , edges = $\frac{n(n-1)}{2}$.

7 Number Theory

Study of integers and their properties.

7.1 Basics

- **Divisibility:** $a|b$ if $b = ka$ for integer k .
- **GCD/LCM:** Greatest common divisor (Euclidean algorithm: $\gcd(a, b) = \gcd(b, a \bmod b)$); $\text{LCM}(a, b) = \frac{ab}{\gcd(a, b)}$.
- **Primes:** Numbers > 1 with no divisors but 1 and itself (Fundamental Theorem: Unique prime factorization).
- **Modular Arithmetic:** $a \equiv b \pmod{m}$ if m divides $(a - b)$.
 - Inverses: Exists if $\gcd(a, m) = 1$.
 - Chinese Remainder Theorem: Solve systems if moduli coprime.

7.2 Advanced

- **Fermat's Little Theorem:** If p prime, $a^{p-1} \equiv 1 \pmod{p}$ (a not multiple of p).
- **Euler's Totient** $\phi(n)$: Count integers $< n$ coprime to n .

Example: $\gcd(48, 18)$: $48 = 2 \cdot 18 + 12$, $18 = 1 \cdot 12 + 6$, $12 = 2 \cdot 6 + 0 \rightarrow \gcd = 6$.

8 Discrete Probability

Probability on discrete spaces.

8.1 Basics

- **Sample Space:** All outcomes.
- **Events:** Subsets; $P(A) = \frac{|A|}{|\text{sample}|}$ for uniform.
- **Conditional** $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
- **Independence:** $P(A \cap B) = P(A)P(B)$.
- **Bayes' Theorem:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$.

8.2 Random Variables

- **Discrete RV:** Maps outcomes to numbers (e.g., dice roll).
- **Expectation** $E[X] = \sum xP(X = x)$.
- **Variance** $\text{Var}(X) = E[(X - E[X])^2]$.
- **Distributions:** Binomial (n trials, p success: $P(k) = C(n, k)p^k(1 - p)^{n-k}$), Poisson (rare events).

Example: Coin flip (fair): $P(\text{heads}) = \frac{1}{2}$, $E[\text{heads in 2 flips}] = 1$.

9 Boolean Algebra and Circuits

- **Boolean Variables:** True/False.
- **Operations:** AND, OR, NOT (like $\wedge$, $\vee$, $\neg$).
- **Laws:** Idempotent, distributive, etc.
- **Applications:** Logic gates in computers, Karnaugh maps for simplification.

10 Automata and Computability

- **Finite Automata:** Machines for regular languages (DFA/NFA).
- **Turing Machines:** Model computation; halting problem undecidable.
- **Complexity:** P vs. NP (open problem).

11 Applications in Computer Science

- **Algorithms:** Graph algorithms for networks.
- **Databases:** Relations as sets.
- **Cryptography:** RSA uses modular inverses.
- **AI:** Logic for reasoning, combinatorics for search spaces.
- **Software Engineering:** Proofs for correctness.

12 Best Practices and Learning Tips

- **Practice Proofs:** Start with simple inductions.
- **Visualize:** Draw graphs, sets.
- **Tools:** Use Python/SageMath for computations (e.g., graph libraries).
- **Common Pitfalls:** Off-by-one in counting, assuming continuity.